MATHS MANTHAN
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 05
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. If p and q are two co-prime natural numbers, then LCM(p, q) is equal to: [1]

(a) 1 (b) $\frac{p}{q}$ (c) $p + q$ (d) pq

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (d) pq

Since p and q are co-prime, HCF$(p, q) = 1$. We know:

Formula Used: $\text{HCF}(p, q) \times \text{LCM}(p, q) = p \times q$

$1 \times \text{LCM}(p, q) = pq \implies \text{LCM}(p, q) = \boxed{pq}$. [1M]

Q2. Given that HCF$(306, 657) = 9$, the value of LCM$(306, 657)$ is: [1]

(a) 22338 (b) 21338 (c) 22334 (d) 23238

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 22338

We know the product relation:

Formula Used: $\text{LCM}(a, b) = \frac{a \times b}{\text{HCF}(a, b)}$

$\text{LCM}(306, 657) = \frac{306 \times 657}{9} = 34 \times 657 = \boxed{22338}$. [1M]

Q3. Given that $\text{LCM}(91, 26) = 182$, the value of $\text{HCF}(91, 26)$ is: [1]

(a) 26 (b) 13 (c) 7 (d) 9

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 13

We know the identity connecting HCF and LCM:

Formula Used: $\text{HCF}(a, b) = \dfrac{a \times b}{\text{LCM}(a, b)}$

$\text{HCF}(91, 26) = \dfrac{91 \times 26}{182} = \dfrac{91 \times 1}{7} = \boxed{13}$. [1M]

Q4. Which of the following expressions represents a rational number? [1]

(a) $(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})$ (b) $(2 + \sqrt{3})^2$

(c) $\dfrac{\sqrt{3}}{2}$ (d) $3\sqrt{2}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) $(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})$

We know the algebraic difference-of-squares identity:

Formula Used: $(a - b)(a + b) = a^2 - b^2$

Evaluating (a): $(\sqrt{2}-\sqrt{3})(\sqrt{2}+\sqrt{3}) = (\sqrt{2})^2 - (\sqrt{3})^2 = 2 - 3 = \boxed{-1 \in \mathbb{Q}}$. Meanwhile (b) $= 7 + 4\sqrt{3}$, (c) $= \frac{\sqrt{3}}{2}$, and (d) $= 3\sqrt{2}$ are all irrational. [1M]

Q5. **Assertion (A):** The HCF and LCM of two natural numbers can be 18 and 380 respectively.
Reason (R): The LCM of two natural numbers is always completely divisible by their HCF. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (D) (A) is false but (R) is true.

We know the fundamental divisibility condition between HCF and LCM:

Formula Used: For any positive integers a, b : $\text{HCF}(a, b) \mid \text{LCM}(a, b)$

Dividing 380 by 18: $380 = 18 \times 21 + 2$, leaving remainder $2 \neq 0$ (not divisible). Therefore (A) is false while Reason (R) is true. [1M]

SECTION B: Very Short Answer Type

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Q6. Find the HCF and LCM of 612 and 1314 using the prime factorisation method. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Prime factorising 612 and 1314:

$$612 = 4 \times 9 \times 17 = 2^2 \times 3^2 \times 17, \qquad 1314 = 2 \times 9 \times 73 = 2^1 \times 3^2 \times 73. \qquad [1M]$$

We know the prime-power HCF and LCM rules:

Formula Used: $\text{HCF} = \prod p^{\min \text{ exp}}, \qquad \text{LCM} = \prod p^{\max \text{ exp}}$

Common primes are 2 and 3 with smallest powers $2^1, 3^2$:

$$\text{HCF}(612, 1314) = 2 \times 3^2 = \boxed{18}.$$

Taking greatest powers of all primes $2, 3, 17, 73$:

$$\text{LCM}(612, 1314) = 2^2 \times 3^2 \times 17 \times 73 = 36 \times 1241 = \boxed{44676}. \qquad [1M]$$

SECTION C: Short Answer Type

MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Three sets of English, Hindi, and Mathematics books have to be stacked in such a way that all the books are stored topic-wise and the height of each stack is the same. The number of English books is 96, Hindi books is 240, and Mathematics books is 336. Assuming the books are of identical thickness, determine the number of books in each stack and the total number of stacks. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Equal stack height with identical-thickness books requires $\text{HCF}(96, 240, 336)$ books per stack. We know:

Formula Used: $\text{Books per stack} = \text{HCF}(96, 240, 336), \qquad \text{Total stacks} = \sum \frac{\text{Subject count}}{\text{HCF}}$

Prime factorising:

$$96 = 2^5 \times 3, \qquad 240 = 2^4 \times 3 \times 5, \qquad 336 = 2^4 \times 3 \times 7. \qquad [1M]$$

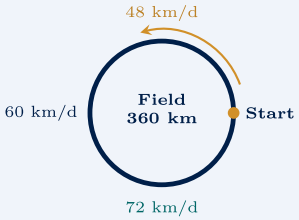
Common primes 2 and 3 have least powers 2^4 and 3^1:

$$\text{HCF}(96, 240, 336) = 2^4 \times 3 = 16 \times 3 = \boxed{48 \text{ books per stack}}. \qquad [1M]$$

Calculating subject stacks: English $= \frac{96}{48} = 2$, Hindi $= \frac{240}{48} = 5$, Mathematics $= \frac{336}{48} = 7$.
Total stacks $= 2 + 5 + 7 = \boxed{14 \text{ stacks}}$. [1M]

Q8. A circular field has a circumference of 360 km. Three cyclists start together and can cycle 48 km, 60 km, and 72 km a day around the field. When will they meet again at the starting point? **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Circumference $C = 360$ km; cyclist speeds $48, 60, 72$ km/day. Each returns to the starting marker after exact integer multiples of their round durations.

Step 1: Time for one full round by each cyclist: [1M]

$$T_1 = \frac{360}{48} = \frac{15}{2} \text{ days } (= 7.5 \text{ days}), \quad T_2 = \frac{360}{60} = 6 \text{ days}, \quad T_3 = \frac{360}{72} = 5 \text{ days}.$$

We know the LCM formula for fractions to reunite simultaneously at the starting point:

Formula Used: $\text{LCM}\left(\frac{a}{p}, \frac{b}{q}, \frac{c}{r}\right) = \frac{\text{LCM}(a,b,c)}{\text{HCF}(p,q,r)}$

Step 2: Computing simultaneous meeting time in days: [2M]

With numerators $15, 6, 5$ and denominators $2, 1, 1$:

$$\text{LCM}\left(\frac{15}{2}, \frac{6}{1}, \frac{5}{1}\right) = \frac{\text{LCM}(15,6,5)}{\text{HCF}(2,1,1)} = \frac{30}{1} = \boxed{30 \text{ days (i.e., } 720 \text{ hours)}}.$$

Check: In 30 days, they cover $4, 5$, and 6 complete rounds respectively ($\text{HCF}(4,5,6) = 1$, confirming earliest reunion).

Q9. Four bells toll together at 9:00 AM. They toll at intervals of $6, 8, 12$, and 18 seconds respectively. How many times will they toll together in the next 1 hour (excluding the toll at 9:00 AM)? **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

The bells toll together again after every interval equal to $\text{LCM}(6, 8, 12, 18)$. We know:

Formula Used: Interval $= \text{LCM}(6, 8, 12, 18)$, Future tolls (excluding start) $= \frac{T_{\text{total}}}{\text{LCM}}$

Prime factorising $6 = 2 \times 3$, $8 = 2^3$, $12 = 2^2 \times 3$, $18 = 2 \times 3^2$:

$$\text{LCM}(6, 8, 12, 18) = 2^3 \times 3^2 = 8 \times 9 = 72 \text{ seconds.} \quad \textbf{[1.5M]}$$

Converting 1 hour $= 60 \times 60 = 3600$ seconds. Excluding the initial 9:00 AM toll:

$$\text{Number of times} = \frac{3600}{72} = \boxed{50 \text{ times}}. \quad \textbf{[1.5M]}$$

Q10. CASE STUDY 05: CONFECTIONERY SWEET SHOP TRAY PACKING [4 Marks]

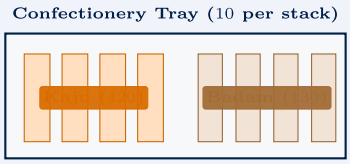
A traditional sweet manufacturer prepares a premier batch of artisanal sweets for a festive order. The batch consists of 420 pieces of Kaju Barfi and 130 pieces of Badam Barfi. The packaging team needs to stack them neatly on display trays such that each stack has the same number of pieces, each stack consists of only one variety of barfi, and the stacks occupy the minimum possible surface area of each tray.

(i) Find the maximum number of barfis that can be placed in each stack to minimize the tray area. [2]

(ii) Calculate the total number of stacks formed on the trays. [1]

(iii) Express 420 in its canonical prime factorisation form. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Area Minimization: Placing the highest common number of sweets (HCF = 10) vertically in each stack reduces the footprint on the display trays to $42 + 13 = 55$ stacks total.

We know the HCF criterion for maximum equal stack height:

Formula Used: Max pieces per stack = HCF(420, 130)

(i) Max barfis per stack: Factorising $420 = 2^2 \times 3 \times 5 \times 7$ and $130 = 2 \times 5 \times 13$:

$$\text{HCF}(420, 130) = 2^1 \times 5^1 = \boxed{10 \text{ barfis per stack}}. \quad [2M]$$

(ii) Total stacks formed:

$$\text{Total stacks} = \frac{420}{10} + \frac{130}{10} = 42 + 13 = \boxed{55 \text{ stacks}}. \quad [1M]$$

(iii) Canonical prime factorisation of 420: $420 = 4 \times 3 \times 5 \times 7 = \boxed{2^2 \times 3 \times 5 \times 7}$. [1M]

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